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Flow to a Partially Penetrating Well in a Double-Porosity Reservoir

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Analysis of flow in a fractured porous reservoir forms the basis for investigations of chemical and energy transport in such media. Numerical models are often employed to analyze these geohydrologic systems. In this paper a well hydraulics problem is solved using the Laplace transformation and the double-porosity concept. The transient solution is obtained by numerical inversion of the Laplace transform. Solutions to slug test problems indicate that the head response due to fracturing is distinct from the response due to partial penetration or skin effect. An alternative to the commonly used van Everdingen model of skin effect is given. No method for identifying fractured porous reservoir parameters from slug tests has been developed. The results of this paper may be applied to test numerical models of flow in fractured porous media.

1. INTRODUCTION

Flow in fractured porous reservoirs is of great interest and importance to hydrologists and petroleum engineers. Current activities have been stimulated by waste disposal studies, water supply needs, and energy-related concerns. The ability to characterize and forecast flow in these media is a basic requirement for attempts to predict transport of wastes or energy.

In this paper a Laplace domain solution of a well hydraulics problem in a fractured porous medium is developed. The so-called double-porosity approach [*Barenblatt et al.*, 1960] is employed, rather than the discrete or enumerative approach in which the geometry of individual fractures is specified. The fundamental double-porosity equations for a liquid-saturated reservoir, given by Barenblatt et al., are two diffusion equations coupled by a mass exchange term. The constitutive relationship for this term is the subject of ongoing debate. The earliest and most widely used expression is the quasi-steady approximation of Barenblatt et al. Other forms of the exchange term exist [*de Swaan*, 1976; *Streltsova-Adams*, 1978; *Shapiro*, 1981; *Huyakorn et al.*, 1983], all of which require some information about the fracture geometry. In a recent paper, *Moench* [1984] has examined a situation which justifies the use of the quasi-steady formulation over more elaborate expressions.

A number of analytic solutions for wellflow problems in confined fractured porous reservoirs have appeared in the literature [e.g., *Barenblatt et al.*, 1960; *Warren and Root*, 1963; *Streltsova-Adams*, 1978; *Da Prat*, 1981], all of which assume full penetration. The case of a partially penetrating well has been examined by *Hantush* [1962] for a single-porosity reservoir of infinite radial extent with constant flow rate. *Agarwal et al.* [1970] considered a fully penetrating constant flow rate well having wellbore storage and skin effect. *Mavor and Cinco-Ley* [1979] studied a similar problem in a double-porosity reservoir. Here these analyses are extended to include all of the above effects along with a non-equilibrium initial condition in the wellbore.

An outline of the paper follows. In section 2 the governing equations along with initial and boundary conditions are presented. They are made dimensionless in section 3. These equations are solved in section 4 in the Laplace transform domain. After some remarks the numerical scheme to invert the solution to the time domain is presented in section 6. Several slug tests are given as applications of the solution in section 7. Closing remarks appear in section 8.

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2. SPECIFICATION OF THE PROBLEM

A homogeneous, isotropic reservoir is confined above and below by parallel, impervious planes separated by a distance b' (primed quantities have dimension). A cylindrical coordinate system is introduced with the z' axis normal to the bounding planes and parallel to gravity and $z' = 0$ at the bottom boundary. The medium is of infinite radial extent, and a vertical wellbore is centered at $r' = 0$. A screen or open interval of length Δ' with radius r_s' exists along the wellbore from $z' = z_a'$ to z_b' and the wellbore is cased above the screened interval. The effective casing radius r_c' is selected such that $\pi r_c'^2$ equals the area of the casing cross section at the liquid surface in the bore, so that storage effects are considered correctly. Both r_s' and r_c' are assumed to be constants. The problem is presented schematically in Figure 1, and notation is presented at the end of the text.

The slightly compressible homogeneous fluid and the solids filling the reservoir volume are considered to be subject to isothermal conditions. Therefore hydraulic head may be introduced and the Darcy law substituted into the mass balances, which can then be written

$$K_f'\left(\frac{\partial^2 h_f'}{\partial r'^2} + \frac{1}{r'}\frac{\partial h_f'}{\partial r'} + \frac{\partial^2 h_f'}{\partial z'^2}\right) = c_f'\frac{\partial h_f'}{\partial t'} - \Gamma' \qquad (1)$$

$$K_m'\left(\frac{\partial^2 h_m'}{\partial r'^2} + \frac{1}{r'}\frac{\partial h_m'}{\partial r'} + \frac{\partial^2 h_m'}{\partial z'^2}\right) = c_m'\frac{\partial h_m'}{\partial t'} + \Gamma' \qquad (2)$$

where K_f' and K_m' are the hydraulic conductivities of the fractures and matrix, respectively, Γ' is the rate of fluid transfer from the matrix to the fractures, and c_f' and c_m' are specific storage coefficients.

Assume that [*Warren and Root*, 1963]

$$K_m' \ll K_f' \qquad (3)$$

and [*Barenblatt et al.*, 1960]

$$\Gamma' = -\lambda'(h_f' - h_m') \qquad (4)$$

where λ' is a coefficient representing the ease of fluid transfer between the fractures and the matrix. Then (2) becomes

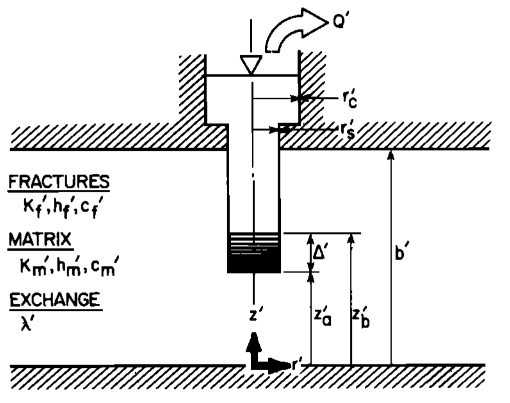

Fig. 1. Schematic of a general double-porosity well hydraulics problem.

$$c_m' \frac{\partial h_m'}{\partial t'} = \lambda'(h_f' - h_m') \tag{5}$$

The wellbore is considered next. A macroscopic mass balance requires that the change in mass in the wellbore (due to a volume or liquid level change plus compressibility) be equal to the total influx through the screen, hereinafter called the total "sand face" flux, minus the efflux Q' from the top of the wellbore. Assuming the vertical pressure gradients in the wellbore to be hydrostatic, the wellbore balance equation is [*van Everdingen and Hurst*, 1949]

$$c_w' \frac{\partial h_w'}{\partial t'} = -Q' + 2\pi r_s' q_{sf}^{*\prime} \Delta' \tag{6}$$

where $2\pi r_s' q_{sf}^{*\prime}\Delta'$ is the total flux through the screen into the wellbore, i.e., $q_{sf}^{*\prime}$ is the average sand face flux:

$$q_{sf}^{*\prime} = \frac{1}{\Delta'} \int_{z_a'}^{z_b'} q_{sf}' \, dz \tag{7}$$

When the fluid compressibility is neglected in the wellbore accumulation terms because volume changes due to liquid level dominate, then

$$c_w' = \pi r_c'^2 \tag{8}$$

(Note that the reservoir equations include both fluid and solid skeleton compressibility, so the terms c_f' and c_m' are not neglected.)

Next, the head in the well h_w' is related to the average head in the reservoir just outside the wellbore in the fractures. We assume that they differ by an amount proportional to the sand face flux, that is,

$$h_w'(t') = h_f^{*\prime}(r_s', t') - S' \, q_{sf}^{*\prime}(t') \tag{9}$$

where S' is the skin (or wellbore damage) factor and $h_f^{*\prime}$ is the average head along the screened section of wellbore:

$$h_f^{*\prime}(r_s', t') = \frac{1}{\Delta'} \int_{z_a'}^{z_b'} h_f'(r_s', z', t') \, dz \tag{10}$$

This skin equation (9) was introduced by *van Everdingen* [1953] and *Hurst* [1953] and will be discussed later in the paper.

Next the initial values and boundary conditions are stated. By virtue of Darcy's law, the radial flow into the wellbore is

$$K_f' \left.\frac{\partial h_f'}{\partial r'}\right|_{r'=r_s'} (z', t') = q_{sf}'(z', t') \qquad z_a' \le z' \le z_b' \tag{11a}$$

$$K_f' \left.\frac{\partial h_f'}{\partial r'}\right|_{r'=r_s'} (z', t') = 0 \qquad z' < z_a' \qquad z' > z_b' \tag{11b}$$

where $q_{sf}'(z', t')$ is the local flux through the sand face into the well. Note the assumption, consistent with the approach of *Warren and Root* [1963], that all flux through the sandface is derived from the fractures. There is no flow through the upper and lower boundary planes and no head changes as $r \to \infty$. We assume that the initial head in the reservoir is constant and that the initial head in the wellbore differs from that of the reservoir by an amount h_{wo}'. That is,

$$h_f'(r', z', 0) = h_m'(r', z', 0) = h_0' \tag{12}$$

$$h_w'(0) = h_0' + h_{wo}' \tag{13}$$

$$h_f'(r', z', t') = h_m(r', z', t') = h_0' \qquad r' \to \infty \tag{14}$$

$$\frac{\partial h_f'(r', z', \tau')}{\partial z'} = 0 \qquad z' = 0, b' \tag{15}$$

Because the head in the wellbore is assumed to vary in time only, it is reasonable to assume that h_f' shows no dependence on z' along the sandface, $r' = r_s'$. Henceforth we assume that q_{sf}' is independent of z' and varies only with time. Its unknown magnitude will be obtained as part of the solution procedure. For all values of z' outside of the open interval $z_a' \le z' \le z_b'$ at $r' = r_s'$ we apply the zero radial flux condition. The analysis therefore assumes that the borehole is cased at radius r_s' above and below the open interval through the reservoir thickness b'. This completes the fomulation of the problem.

3. DIMENSIONLESS VARIABLES

The problem is next cast into dimensionless (unprimed quantities) form. The dimensionless groups are:

$$t = \frac{K_f' t'}{(c_f' + c_m') r_s'^2} \tag{16}$$

$$r = r'/r_s' \tag{17}$$

$$z = z'/r_s' \tag{18}$$

$$h_\alpha = \frac{2\pi K_f' \Delta'}{Q_0'} (h_0' - h_\alpha') \qquad \alpha = f, m, w \tag{19}$$

$$Q = Q'/Q_0' \tag{20}$$

$$q_{sf} = \frac{2\pi r_s'^2 \Delta'}{Q_0'} q_{sf}' \tag{21}$$

$$\Delta = \Delta'/r_s' \tag{22}$$

$$\rho = \Delta'/b' \tag{23}$$

Here Q is the normalized efflux and ρ is the penetration ratio. Cases of constant efflux are represented by $Q = 1$. The quantity q_{sf} is a time-varying dimensionless flux.

The reservoir and wellbore properties are characterized by four subsidiary dimensionless groups:

$$\lambda = \frac{\lambda' r_s'^2}{K_f'} \tag{24}$$

$$C_D = \frac{c_w'}{2\pi r_s'^2 (c_f' + c_m') \Delta'} \tag{25}$$

$$S = \frac{S' K_f'}{r_s'} \tag{26}$$

$$\omega = \frac{c_f'}{c_f' + c_m'} \tag{27}$$

It is through these dimensionless parameters that the solution may be simplified to some well-known solutions of well hydraulics. For example, if $\omega = 1$, then c_m' must be zero, so the fracture-matrix exchange term equals zero and (1) is left in the standard, single-porosity form.

The governing equations in dimensionless form read:

$$\frac{\partial^2 h_f}{\partial r^2} + \frac{1}{r}\frac{\partial h_f}{\partial r} + \frac{\partial^2 h_f}{\partial z^2} = \omega \frac{\partial h_f}{\partial t} + (1-\omega)\frac{\partial h_m}{\partial t} \tag{28}$$

$$(1-\omega)\frac{\partial h_m}{\partial t} = \lambda(h_f - h_m) \tag{29}$$

$$Q = C_D \frac{\partial h_w}{\partial t} + q_{sf} \tag{30}$$

$$h_w = h_f^* + S q_{sf} \tag{31}$$

The average head h_f^* is defined as

$$h_f^* = \frac{1}{\Delta}\int_{z_a}^{z_b} h_f|_{r=1}\, dz \tag{32}$$

The initial and boundary conditions are

$$h_f(r, z, t) = h_m(r, z, t) = 0 \qquad t = 0 \tag{33}$$

$$h_w(t) = h_{w_0} \qquad t = 0 \tag{34}$$

$$\partial h_f/\partial r\,|_{r=1} = -q_{sf} \qquad z_a \le z \le z_b \tag{35a}$$

$$\partial h_f/\partial r\,|_{r=1} = 0 \qquad z < z_a \qquad z > z_b \tag{35b}$$

$$h_f(r, z, t) = h_m(r, z, t) = 0 \qquad r \to \infty \tag{36}$$

$$\partial h_f/\partial z = 0 \qquad z = 0, b \tag{37}$$

where $b = b'/r_s'$.

4. Solution of the Problem

The solution is obtained by means of the Laplace transformation. Let the Laplace transform of a function $f(t)$ be denoted by $\bar{f}(\beta)$, where β is the transform variable. Equations (28)–(37) are now given by

$$\frac{\partial^2 \bar{h}_f}{\partial r^2} + \frac{1}{r}\frac{\partial \bar{h}_f}{\partial r} + \frac{\partial^2 \bar{h}_f}{\partial z^2} = \omega\beta\bar{h}_f + (1-\omega)\beta\bar{h}_m \tag{38}$$

$$(1-\omega)\beta\bar{h}_m = \lambda(\bar{h}_f - \bar{h}_m) \tag{39}$$

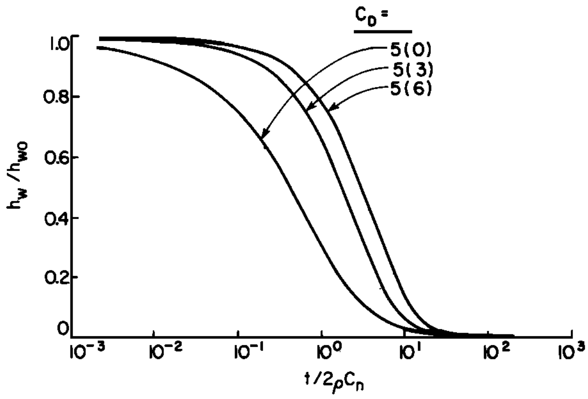

Fig. 2. Slug test curves. *Cooper et al.*'s [1967] results for the fully penetrating, single-porosity, no-skin case.

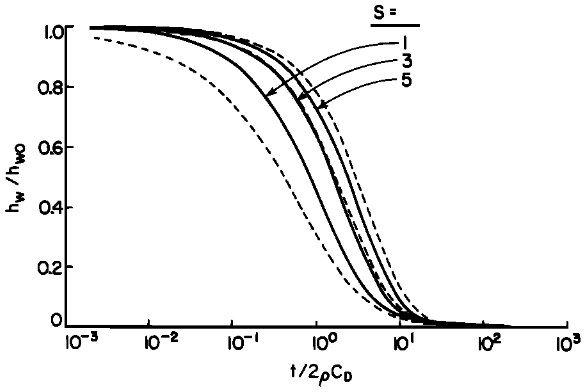

Fig. 3. Slug test curves. Effect of various values of the van Everdingen skin factor (solid lines) along with the Cooper et al. curves (dashed lines).

$$\bar{Q} = \beta C_D \bar{h}_w - C_D h_{w_0} + \bar{q}_{sf} \tag{40}$$

$$\bar{h}_w = \bar{h}_f^* + S\bar{q}_{sf} \tag{41}$$

$$\bar{h}_f^* = \frac{1}{\Delta}\int_{z_a}^{z_b} \bar{h}_f|_{r=1}\, dz \tag{42}$$

$$\partial \bar{h}_f/\partial r\,|_{r=1} = -\bar{q}_{sf} \qquad z_a \le z \le z_b \tag{43a}$$

$$\partial \bar{h}_f/\partial r\,|_{r=1} = 0 \qquad z < z_a \qquad z > z_b \tag{43b}$$

$$\bar{h}_f = 0 \qquad r \to \infty \tag{44}$$

$$\partial \bar{h}_f/\partial z = 0 \qquad z = 0, b \tag{45}$$

Rearranging (39) yields

$$\bar{h}_m = \frac{\lambda \bar{h}_f}{(1-\omega)\beta + \lambda} \tag{46}$$

Equations (46) and (38) are combined to form

$$\frac{\partial^2 \bar{h}_f}{\partial r^2} + \frac{1}{r}\frac{\partial \bar{h}_f}{\partial r} + \frac{\partial^2 \bar{h}_f}{\partial z^2} = \beta f \bar{h}_f \tag{47}$$

where

$$f = \frac{\omega(1-\omega)\beta + \lambda}{(1-\omega)\beta + \lambda} \tag{48}$$

We seek a solution of (47) satisfying (45) in the form

$$\bar{h}_f = \sum_{n=0}^{\infty} \bar{g}_n(r, \beta) \cos\left(\frac{n\pi z}{b}\right) \tag{49}$$

Substitution of (49) into (47) yields:

$$\sum_{n=0}^{\infty}\left[\bar{g}_n'' + \frac{1}{r}\bar{g}_n' - \left(\beta f + \frac{n^2\pi^2}{b^2}\right)\bar{g}_n\right]\cos\left(\frac{n\pi z}{b}\right) = 0 \tag{50}$$

where primes denote differentiation with respect to r, leading to

$$\bar{g}_n'' + \frac{1}{r}\bar{g}_n' - \left(\beta f + \frac{n^2\pi^2}{b^2}\right)\bar{g}_n = 0 \qquad n = 0, 1, 2, \cdots \tag{51}$$

The boundedness condition (44) gives

$$\bar{g}_n = \alpha_n(\beta) K_0(q_n r) \qquad n = 0, 1, 2, \cdots \tag{52}$$

where α_n is a function of β only, K_0 is the modified Bessel function of the second kind of order zero, and

$$q_n = \left(\beta f + \frac{n^2\pi^2}{b^2}\right)^{1/2} \tag{53}$$

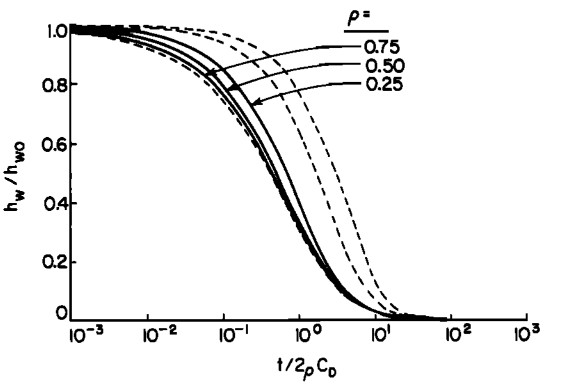

Fig. 4. Slug test curves. Effect of various degrees of penetration (solid lines) along with the Cooper et al. curves (dashed lines).

Thus the solution (49) of (47) is given by

$$\bar{h}_f = \alpha_0 K_0(q_0 r) + \sum_{n=1}^{\infty} \alpha_n K_0(q_n r) \cos\left(\frac{n\pi z}{b}\right) \tag{54}$$

Next the Fourier cosine coefficients α_n in (54) are determined from (43):

$$\alpha_0 q_0 K_1(q_0) + \sum_{n=1}^{\infty} \alpha_n q_n K_1(q_n) \cos(n\pi z/b) = \bar{q}_{sf} \qquad z_a \le z \le z_b \tag{55a}$$

$$\alpha_0 q_0 K_1(q_0) + \sum_{n=1}^{\infty} \alpha_n q_n K_1(q_n) \cos(n\pi z/b) = 0 \qquad z < z_a \quad z > z_b \tag{55b}$$

leading to

$$\alpha_0 = \frac{\Delta \bar{q}_{sf}}{b q_0 K_1(q_0)} = \frac{\rho \bar{q}_{sf}}{q_0 K_1(q_0)} \tag{56a}$$

$$\alpha_n = \frac{2\bar{q}_{sf}}{n\pi q_n K_1(q_n)}\left[\sin\frac{n\pi z_b}{b} - \sin\frac{n\pi z_a}{b}\right] \qquad n = 1, 2, \cdots \tag{56b}$$

When the expressions for α_0 and α_n are substituted into (54), we find the head in the fractures within the reservoir:

$$\bar{h}_f(r, z, \beta) = \rho \bar{q}_{sf}\left\{\frac{K_0(q_0 r)}{q_0 K_1(q_0)} + \frac{2}{\rho}\sum_{n=1}^{\infty}\left[K_0(q_n r)\left(\sin\frac{n\pi z_b}{b} - \sin\frac{n\pi z_a}{b}\right)\cos\frac{n\pi z}{b}\,[n\pi q_n K_1(q_n)]^{-1}\right]\right\} \tag{57}$$

The skin equation (41) enables calculation of the wellbore head from the head in the reservoir. First, we compute $\bar{h}_f{}^*$

$$\bar{h}_f{}^* = \frac{1}{\Delta}\int_{z_a}^{z_b} \bar{h}_f|_{r=1}\, dz = \frac{1}{\Delta}\int_{z_a}^{z_b} \frac{\rho \bar{q}_{sf}}{q_0 K_1(q_0)} K_0(q_0)\, dz + \sum_{n=1}^{\infty}\frac{1}{\Delta}\int_{z_a}^{z_b}\frac{2\bar{q}_{sf}}{n\pi q_n K_1(q_n)} \cdot \left[\sin\frac{n\pi z_b}{b} - \sin\frac{n\pi z_a}{b}\right] K_0(q_n)\cos\frac{n\pi z}{b}\, dz$$

$$= \rho \bar{q}_{sf}\left\{\frac{K_0(q_0)}{q_0 K_1(q_0)} + \frac{2}{\rho^2}\sum_{n=1}^{\infty}\left[\left(\sin\frac{n\pi z_b}{b} - \sin\frac{n\pi z_a}{b}\right)^2 K_0(q_n) \cdot [n^2\pi^2 q_n K_1(q_n)]^{-1}\right]\right\} \tag{58}$$

Next, we substitute (58) into (41), finding

$$\bar{h}_w(\beta) = \rho \bar{q}_{sf}\left\{\frac{K_0(q_0)}{q_0 K_1(q_0)} + \frac{2}{\rho^2}\sum_{n=1}^{\infty}\left[\left(\sin\frac{n\pi z_b}{b} - \sin\frac{n\pi z_a}{b}\right)^2 K_0(q_n) \cdot [n^2\pi^2 q_n K_1(q_n)]^{-1}\right]\right\} + S\bar{q}_{sf} \tag{59}$$

To simplify matters, some new notation is introduced. We define

$$A_0 = \frac{K_0(q_0)}{q_0 K_1(q_0)} \tag{60a}$$

$$A = \frac{2}{\rho^2}\sum_{n=1}^{\infty}\left[\left(\sin\frac{n\pi z_b}{b} - \sin\frac{n\pi z_a}{b}\right)^2 K_0(q_n) \cdot [n^2\pi^2 q_n K_1(q_n)]^{-1}\right] \tag{60b}$$

$$E_0 = \frac{K_0(q_0 r)}{q_0 K_1(q_0)} \tag{60c}$$

$$E = \frac{2}{\rho}\sum_{n=1}^{\infty}\left[K_0(q_n r)\left(\sin\frac{n\pi z_b}{b} - \sin\frac{n\pi z_a}{b}\right)\cos\frac{n\pi z}{b} \cdot [n\pi q_n K_1(q_n)]^{-1}\right] \tag{60d}$$

From (40) we have

$$\bar{q}_{sf} = \bar{Q} - \beta C_D \bar{h}_w + C_D h_{wo} \tag{61}$$

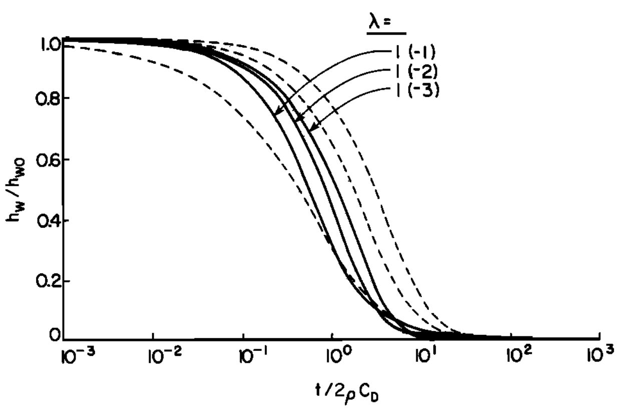

Fig. 5. Slug test curves. Effect of values of the double-porosity exchange coefficient (solid lines) along with the Cooper et al. curves (dashed lines).

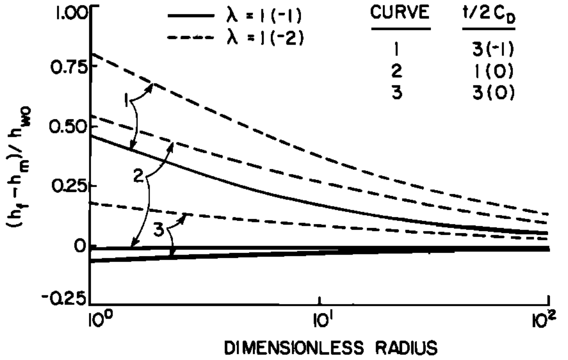

Fig. 6. Slug test curves. Radial distribution of normalized head difference between fractures and matrix for selected values of λ and $t/2C_D$.

Then, after some algebraic simplifications, (57), (59), and (61) yield

$$\bar{h}_w(\beta) = \frac{(\bar{Q} + C_D h_{wo})[\rho(A_0 + A) + S]}{1 + \beta C_D[\rho(A_0 + A) + S]} \tag{62}$$

$$\bar{h}_f(r, z, \beta) = \frac{(\bar{Q} + C_D h_{wo})\rho[E_0 + E]}{1 + \beta C_D[\rho(A_0 + A) + S]} \tag{63}$$

The head in the matrix, $\bar{h}_m(r, z, \beta)$, is given by (46).

Equations (62) and (63) are simplified for some important special cases. A single-porosity solution is obtained by simply setting $\omega = 1$, which alters the arguments of the Bessel functions (53) to

$$q_n = \left(\beta + \frac{n^2\pi^2}{b^2}\right)^{1/2} \tag{64}$$

Another important case is that of full penetration. In this circumstance the terms A and E defined above are identically zero and $\rho = 1$. These and other special cases are indicated in Table 1.

5. REMARKS

The results of the analysis, (62) and (63), have not been transformed to the time domain by analytical methods. Some asymptotic expressions may be derived as have been presented for many other well hydraulics problems [for example, *Agarwal et al.*, 1970]. To ease such restrictions on the results that we obtain, a numerical inversion procedure has been applied and is discussed in ensuing sections.

The major simplifications introduced in the analysis, after the assumption of the Warren and Root quasi-steady equations, are

1. The head in the well is independent of depth.
2. Flux through the screen is independent of depth.
3. The skin effect can be modeled in the fashion described by (8).

Simplification 1 is critical to posing a solvable problem. It would not be easy to relax this assumption and obtain an analytical solution, although any solutions obtained from such a model would be very powerful. Simplification 2 may be modified to allow any distribution in a straightforward manner and will only effect the Fourier coefficients α_n. Simplification 3 is useful and widely used [e.g., *Agarwal et al.*, 1970]. A useful alternative approach to include skin effects is discussed in section 7.

6. STEHFEST ALGORITHM

The *Stehfest* [1970] algorithm has been applied successfully to many reservoir problems over the past several years [for example, *Moench and Ogata*, 1981; *Da Prat*, 1981]. The description of the algorithm by *Moench and Ogata* [1981] is quite clear, and the dissertation of *Da Prat* [1981] provides a FORTRAN listing of a typical application. The algorithm is suitable when the solution in the time domain is smooth and monotonic. Superposition of solutions may be employed for extending investigations (e.g., for buildup or recovery tests).

Concisely, the Stehfest algorithm leads to the approximation

$$f(t) = \mathscr{L}^{-1}[\bar{f}(\beta)] \approx \frac{\ln 2}{t} \sum_{i=1}^{N_{\text{steh}}} W_i \bar{f}\left(\frac{i \ln 2}{t}\right) \tag{65}$$

where $\mathscr{L}^{-1}$ is the inverse Laplace transformation, β is the transform variable, $\bar{f}$ is the solution in the Laplace domain, t is time, W_i is a weight dependent only on N_{steh}, and N_{steh} is the number of Stehfest points in the approximation. N_{steh} is an even number which must be determined for each problem and each computer; a value of 18 has been found suitable for most reservoir problems on the IBM 3081 with 64-bit word length.

TABLE 1. Special Cases of the Laplace Domain Solutions (62) and (63)

Case	Parameters	$\bar{h}_w$	$\bar{h}_f$	Other
1. Single porosity	$\omega = 1$	...	...	$f = 1$
2. Full penetration	$\rho = 1$	$\frac{(\bar{Q} + C_D h_{wo})(A_0 + S)}{1 + \beta C_D(A_0 + S)}$	$\frac{(\bar{Q} + C_D h_{wo})E_0}{1 + \beta C_D(A_0 + S)}$	
3. Full penetration Perfect skin	$\rho = 1$ $S = 0$	$\frac{(\bar{Q} + C_D h_{wo})A_0}{1 + \beta C_D A_0}$	$\frac{(\bar{Q} + C_D h_{wo})E_0}{1 + \beta C_D A_0}$	
4. Full penetration No wellbore storage	$\rho = 1$ $C_D = 0$	$\bar{Q}(A_0 + S)$	$\bar{Q}E_0$	
5. Full penetration No wellbore storage Perfect skin	$\rho = 1$ $C_D = 0$ $S = 0$	$\bar{Q}A_0$	$\bar{Q}E_0$	
6. Slug test	$\bar{Q} = 0$ $h_{wo} \neq 0$	$\frac{C_D h_{wo}[\rho(A_0 + A) + S]}{1 + \beta C_D[\rho(A_0 + A) + S]}$	$\frac{C_D h_{wo}\rho(E_0 + E)}{1 + \beta C_D[\rho(A_0 + A) + S]}$	

The weights are determined:

$$W_i = (-1)^{(N_{\text{steh}}/2)+i} \sum_{k=[(i+1)/2]}^{\min(i, N_{\text{steh}}/2)} \frac{k^{N_{\text{steh}}/2}(2k)!}{(N_{\text{steh}}/2-k)!k!(k-1)!(i-k)!(2k-i)!} \tag{66}$$

where the value of k is evaluated by integer arithmetic.

The Stehfest algorithm possesses these features which make it especially attractive for the applications reported in this paper:

1. The computations involve real numbers only. No complex variables are used.
2. The values of the weights, W_i, depend only upon N_{steh}. Therefore they are evaluated once and stored for all subsequent points in time.
3. A small number N_{steh} of evaluations of the Laplace domain solution are required for each point in time.

7. APPLICATION: SLUG TESTS

The nature of the response of a well to an instantaneous charge or discharge of fluid leads to a fast and inexpensive test procedure for fully penetrating wells in confined single-porosity reservoirs. *Cooper et al.* [1967] provide an analytical solution and parameter estimation technique for this test for the case of no skin. Figure 2 is a semilog plot of normalized head in the well, h_w/h_{w0}, versus the dimensionless time $t/2\rho C_D$ for the Cooper et al. problem. The scale of the abscissa has been selected to be identical with that in their paper and, it should be noted, is independent of the length of the open interval. Three wellbore storage constants, $C_D = 5(0)$, 5(3), and 5(6), are shown.

When a skin effect is present the solution is altered as shown in Figure 3, where response of wells with $C_D = 5$ and $S = 1$, 3, and 5 are superposed on the responses given in Figure 2. A shift in the curves to greater values of time with increase in S is clear. It is apparent that use of the Cooper et al. parameter estimation scheme will lead to errors in the estimate of transmissivity. This condition was also observed by *Ramey and Agarwal* [1972]. Replacing the van Everdingen skin equation (8) by the concept of effective screen radius [*Earlougher*, 1977, p. 8]

$$r_{s,\text{effective}}' = r_s' e^{-2S} \tag{67}$$

Ramey et al. [1975] were not only able to develop a correlation function which permits calculation of S but also show that transmissivity estimates by the Cooper et al. method are not sensitive to skin effects.

As a third case, three partially penetrating wells with no skin in a single-porosity reservoir are examined in Figure 4. The parameter values are $\rho = 0.75$, 0.50, 0.25, $b = 1$, $C_D = 5$, and $S = 0$. The bottom of the open interval is at $z = 0$. Usually about 25 terms of the series of (60*b*) are sufficient. Once again the Figure 2 curves are carried along for reference. The penetration ratio ρ appears to have a modest influence on the curves, shifting them up and to the right. However, the dimensionless aquifer thickness b causes greater effects; the curves tend to be pushed down and to the left for larger values of b (not shown here). At early time the response is that of a reservoir with thickness ρb, whereas at large time the full thickness b is involved in the response. This behavior, generated by the numerical inverter, has been verified by an analytic study of the solution.

Finally, we examine a fully penetrating, skin-free well in a double-porosity reservoir. The well storage factor is $C_D = 5$, and $\rho = 1$, S = 0, $\omega = 1(-2)$. The exchange parameter λ is set equal to $1(-1)$, $1(-2)$, and $1(-3)$. At very early time the response is that of a single-porosity reservoir with wellbore storage of C_D/ω. At very late time the solution corresponds to the single-porosity case with wellbore storage of C_D. The transition between the two behaviors is governed by λ.

When λ is large the characteristic time of the matrix response is small compared to the duration of the overall response, so the behavior is essentially that of a single-porosity reservoir with wellbore storage of C_D. When λ is small the characteristic matrix response time is large compared to the duration of the head pulse; that is, the matrix responds very sluggishly to head changes in the fractures. In this case the response is essentially that of a single-porosity reservoir with wellbore storage constant of C_D/ω. At any given radial point, the head in the fractures will first increase then decrease as the slug is dissipated. Examination of the difference between the head in the fractures and the head in the matrix will demonstrate the effect of λ described above. In Figure 6 the normalized head difference is plotted versus radial distance for the $\lambda = 1(-1)$ and $1(-2)$ cases of Figure 5. The differences are shown for three different times, $t/2C_D$ of $3(-1)$, 1(0), and 3(0). The $\lambda = 1(-1)$ curves clearly show that the matrix head lags the fracture head and comparison with the $\lambda = 1(-2)$ curves verifies that the degree of lag increases as λ decreases. Figure 6 also demonstrates that head changes due to the slug test may penetrate radially for over 100 well radii.

8. CONCLUDING REMARKS

The solution presented, in conjunction with the Stehfest algorithm, provides a flexible and inexpensive tool to explore many well hydraulics problems. As a result, numerical models of flow in double-porosity reservoirs can be validated to a degree not before possible. Modification of the solution to describe a reservoir of finite radius or with an alternative specified sandface flux distribution is straightforward.

Several extensions of this work are possible and warranted. Recalling that the delayed yield solution for water table wells has a form similar to the Warren and Root solution [*Streltsova-Adams*, 1978], one immediately suspects that slug test responses in water table single-porosity reservoirs will be similar to those in confined double-porosity reservoirs. This bears investigation. Also, it appears likely that the parameters ω and λ can be identified from slug tests in fully penetrating wells with no skin for certain ranges of λ.

NOTATION

- A group defined by (60*b*).
- A_0 group defined by (60*a*).
- b aquifer thickness.
- c' specific storage.
- C_D dimensionless wellbore storage constant.
- E group defined by (60*d*).
- E_0 group defined by (60*c*).
- f coefficient defined by (48).
- $\bar{g}_n$ nth function given in (49).
- K' hydraulic conductivity.
- K_0, K_1 modified Bessel function of second kind of order 0, 1.
- $\mathscr{L}$, $\mathscr{L}^{-1}$ Laplace transform and its inverse.
- N_{steh} number of points used in Stehfest algorithm.
- Q pumping rate from wellbore.

q_n nth constant given by (53).
q_{sf} radial sandface flux at $r = r_s$.
r radial coordinate.
S skin factor.
t time.
W_i ith weight in Stehfest algorithm.
z vertical coordinate.
α_n nth Fourier coefficient given by (52).
β Laplace transform variable.
Γ mass exchange from matrix to fractures.
λ mass exchange coefficient.
ρ ratio of screen length to aquifer thickness.
ω ratio of fracture to bulk compressibilities.

Subscripts

a bottom of screen.
b top of screen.
c casing at free surface.
f fractures.
m matrix.
s screen.
sf sand face or open interval.
w well.
α dummy variable.

Superscripts

' dimensional quantity (unprimed variables are dimensionless).
* averaged over the vertical.

Other

$\bar{f}$ Laplace transform of f.
5(2) 5×10^2.

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